EXPERNETIC

Expernetic (Pvt) Ltd

Talent Assessment Program · Technical Assignment

Data Engineer Intern

Assignment

ROLE	**Data Engineer Intern**
ASSIGNMENT	**Real-World Data Challenge**
DURATION	**1 Week**
DATASET	**Inside Airbnb (Public)**
FORMAT	**DOCX → Candidate Copy**

🚀 Welcome to the Expernetic Data Challenge

Congratulations on advancing to the technical assessment stage. This assignment is your opportunity to demonstrate analytical thinking, engineering acumen, and your unique approach to solving real-world data problems. There are no trick questions; only open-ended challenges designed to let you shine.

Read the entire document before you begin. Prioritize thoughtfully. Quality outweighs quantity.

01 Introduction

You have been engaged as a Data Engineer and Analyst supporting the business intelligence team of a hypothetical Airbnb market intelligence consultancy. Your stakeholders, including product managers, revenue strategists, and operations leads, need actionable insights derived from publicly available short-term rental data.

The Inside Airbnb dataset provides a rich, real-world snapshot of Airbnb listings, host behaviors, pricing dynamics, availability calendars, and guest reviews across cities worldwide. Your task is to transform this raw data into meaningful engineering artifacts, analytical insights, and business recommendations.

This assignment is deliberately designed to span a wide range of disciplines, from foundational data engineering through to advanced machine learning and generative AI experimentation. The breadth reflects the reality of modern data roles, where practitioners must fluidly move between engineering, analytics, statistics, and applied AI.

💡 Design Philosophy

This assignment is intentionally scoped beyond what can reasonably be completed in one week. That is by design. We are evaluating how you prioritize, how deeply you reason about the problems you choose, and how clearly you communicate your approach, not whether you complete every bullet point.

A candidate who completes two sections with exceptional depth, clean code, and insightful storytelling will outperform one who attempts all sections superficially.

1.1 What This Assignment Tests

Competency	What We Evaluate
Data Engineering Fundamentals	Ingestion, profiling, cleaning, modeling, pipeline design
Analytical Thinking	EDA, statistical reasoning, business interpretation
Data Science & ML	Feature engineering, modeling, validation, explainability
AI & LLM Literacy	Responsible, creative, and effective use of AI tools
Communication & Storytelling	Translating technical findings into business language
Creativity & Initiative	Going beyond the prompt; building something novel
Professionalism	Code quality, documentation, reproducibility, structure

02 Dataset Familiarization

Difficulty	Estimated Effort	Category
🟢 Beginner	2–4 Hours	Mandatory

Before building any pipeline or model, a professional data practitioner always understands the source data thoroughly. This section asks you to invest time in genuine exploration and documentation of the Inside Airbnb dataset.

2.1 Dataset Source

All data for this assignment is sourced exclusively from Inside Airbnb, a publicly available, independent dataset that provides detailed information on Airbnb listings across cities worldwide. No supplementary files or external data will be provided.

- 🔗 **Inside Airbnb Homepage:** https://insideairbnb.com/ — Project overview and methodology
- 🔗 **Dataset Download Page:** https://insideairbnb.com/get-the-data/ — City-level data files
- 🔗 **Data Dictionary:** https://docs.google.com/spreadsheets/d/1iWCNJcSutYqpULSQHlNyGInUvHg2BoUGoNRIGa6Szc4/ — Column definitions for all files

- 🔗 **Behind the Data:** https://insideairbnb.com/behind-the-data/ — Data collection methodology and limitations
- 🔗 **Policy Analysis Tool:** https://insideairbnb.com/research/ — Research and analysis context

2.2 Available Files Per City

Each city dataset typically includes the following files. Availability may vary by city:

File	Description
listings.csv.gz	Core listing data: price, room type, location, host info, amenities
listings.csv (detailed)	Expanded listing attributes including descriptions and full metadata
calendar.csv.gz	Daily price and availability data per listing (365 days)
reviews.csv.gz	Guest review text with reviewer ID, date, and listing reference
reviews.csv (summary)	Aggregated review metrics per listing
neighbourhoods.csv	Neighbourhood names and geospatial groupings
neighbourhoods.geojson	GeoJSON polygon data for neighbourhood boundaries

2.3 Exploration Tasks

- Download at least one city dataset and extract all available files.
- Document the schema of each file: column names, data types, ranges, and sample values.
- Map the relationships between files and identify primary and foreign keys where applicable.
- Review the official data dictionary and note any columns requiring special interpretation.
- Identify dataset limitations: coverage gaps, scraping artifacts, missing historical data.
- Document your assumptions about ambiguous fields before proceeding to analysis.
- Describe the business domain context: what does each entity (listing, host, review) represent?
- If selecting multiple cities: compare schemas across cities and note structural differences.

> 🌍 **Multi-City Guidance**
>
> Candidates selecting multiple cities should additionally: • Compare dataset sizes, recency, and coverage across cities. • Note structural inconsistencies (column names, encodings, formats). • Document your harmonization strategy before engineering begins.

03 Data Engineering Challenges

Difficulty	Estimated Effort	Category
🔵 Intermediate → 🟣 Expert	8–20 Hours	Recommended

Data engineering is at the core of every reliable analytics and machine learning system. This section challenges you to build production-quality data infrastructure, from raw data ingestion through to clean, validated, and analytics-ready datasets.

3.1 Data Ingestion & Profiling

- Design and implement a repeatable ingestion pipeline that downloads and extracts city data.
- Profile each dataset: row counts, column cardinality, null rates, data type distributions.
- Generate a data quality report summarizing findings across all ingested files.
- Detect and document duplicate records using deterministic and fuzzy matching techniques.
- Assess completeness: which fields are most frequently missing and what are the implications?
- Identify outliers in key numerical fields (price, availability, number of reviews).
- Validate data against domain expectations (e.g., price cannot be negative, latitude/longitude must be valid).

3.2 Data Cleaning & Standardization

- Standardize price columns: remove currency symbols, handle thousands separators, cast to numeric.
- Parse and standardize date fields across all relevant files.
- Normalize free-text fields (e.g., property types, room types) into consistent categorical values.
- Handle missing values with appropriate strategies: imputation, sentinel values, or explicit nulls.
- Remove or flag records that fail validation rules, documenting your decisions.
- Standardize geographic fields: neighbourhood names, city names, and coordinate precision.

3.3 Data Enrichment & Joining

- Join listings with review summaries to create an enriched listing master table.
- Integrate calendar data to compute per-listing occupancy rates and revenue estimates.
- Enrich listings with neighbourhood-level aggregates (median price, listing density, average rating).

- Derive calculated fields: host tenure (years on platform), review frequency, price-per-bedroom.
- If multiple cities are selected: build a unified cross-city master dataset with consistent schemas.

3.4 Data Modeling

- Design a dimensional model (star schema) suitable for analytical queries on this dataset.
- Define fact tables and dimension tables; document primary and foreign key relationships.
- Implement your model in a database or analytical engine of your choice (e.g., DuckDB, SQLite, PostgreSQL, BigQuery).
- Write SQL queries demonstrating key analytical use cases against your model.
- Discuss trade-offs in your modeling decisions (denormalization, slowly changing dimensions, etc.).

3.5 Pipeline Design & Automation

- Implement an automated, configurable pipeline that can process any city with minimal code changes.
- Add logging, error handling, and retry logic to your pipeline.
- Implement incremental processing logic and describe how newly scraped data would be handled without requiring full reprocessing.
- Design a metadata management layer: track when datasets were ingested, processed, and validated.
- Document data lineage: for each output table, trace the transformations from source to sink.

3.6 Advanced & Cloud-Native Topics

Difficulty	Estimated Effort	Category
🟣 Expert	6–12 Hours	Optional

- Design a scalable architecture for processing 50+ cities concurrently. What changes?
- Discuss or implement partitioning strategies for the calendar dataset at scale.
- Implement or describe a change data capture (CDC) strategy for the listings table.
- Deploy or describe a cloud-native implementation (AWS, GCP, Azure) with managed services.
- Containerize your pipeline using Docker; provide a working docker-compose.yml.
- Implement data quality checks as code (e.g., Great Expectations, dbt tests, custom framework).
- Discuss production-readiness considerations: monitoring, alerting, SLA tracking, cost optimization.

> 📋 **Engineering Decision Log**
>
> For every significant engineering decision, including tool selection, architectural choice, and schema design, include a brief Decision Log entry in your report explaining: • What options you considered • Why you chose your approach • What trade-offs you accepted. This is where senior engineers and interviewers will focus most of their attention.

04 Exploratory Data Analysis

Difficulty	Estimated Effort	Category
Intermediate	6–10 Hours	Recommended

Exploratory Data Analysis (EDA) is the foundation of any successful analytics project. This section asks you to go beyond summary statistics and uncover the stories hidden within the data, stories that translate directly into business decisions.

4.1 Summary Statistics & Distributions

- Compute and interpret descriptive statistics for all key numerical variables.
- Visualize price distributions by city, neighbourhood, room type, and property type.
- Analyze the distribution of listing counts per host (power law dynamics in hosting).
- Examine review score distributions and identify potential rating inflation patterns.
- Explore availability patterns: how does average availability vary across cities or seasons?

4.2 Geographic & Spatial Analysis

- Map listing density across neighbourhoods using geospatial visualization.
- Identify geographic pricing gradients and determine whether listings closer to city centres command premium prices.
- Visualize review scores spatially to identify neighbourhoods with consistently high or low ratings.
- Explore the geographic clustering of different property and room types.
- If multiple cities: build a comparative geographic overview across markets.

4.3 Temporal & Seasonal Trends

- Analyze how pricing evolves over the calendar year and identify peak seasons.
- Examine review volume trends: are bookings growing or declining in the selected city?
- Explore host tenure distributions: are newer hosts priced differently than established ones?
- Investigate minimum night policies and how they shift across seasons or events.

4.4 Host & Supply-Side Analysis

- Segment hosts by portfolio size (single listing vs. multi-listing commercial operators).
- Compare pricing strategies of professional hosts vs. casual hosts.
- Analyze the relationship between host response rate/superhost status and listing performance.
- Explore the concentration of market supply: what percentage of hosts control the majority of listings?

4.5 Review & Demand-Side Analysis

- Examine the relationship between review count, review score, and price.
- Analyze review frequency as a proxy for booking demand.
- Identify listings with high review counts but low scores and determine what characterizes them.
- Explore review score sub-dimensions (cleanliness, location, communication) if available.

> 📊 **Business Interpretation Requirement**
>
> Every visualization or statistical finding must be accompanied by a plain-English business interpretation. Do not stop at statements such as “the price distribution is right-skewed.” Instead, explain what the finding means for a market participant, platform operator, or investor, and describe the actions it might motivate.

05 Statistical Analysis

Difficulty	Estimated Effort	Category
🔵 Intermediate → 🟠 Advanced	6–12 Hours	Recommended

Statistical rigor separates credible analytical work from surface-level observation. This section challenges you to apply formal statistical methods to substantiate or refute hypotheses about the Airbnb marketplace.

5.1 Hypothesis Testing

- **H1:** Entire-home listings command statistically significantly higher prices than private rooms.
- **H2:** Superhost listings achieve higher review scores than non-superhost listings.
- **H3:** Listings with more than 10 reviews have significantly different prices than listings with fewer.
- **H4:** Neighbourhood average prices differ significantly (ANOVA across neighbourhood groups).
- **H5:** Weekend vs. weekday pricing differences are statistically significant (from calendar data).

For each hypothesis: state the null and alternative, select an appropriate test, check assumptions, report the test statistic, p-value, and effect size, and interpret the result in business terms.

5.2 Confidence Intervals & Effect Sizes

- Compute confidence intervals for mean prices by neighbourhood and room type.
- Report effect sizes (Cohen's d, eta-squared) alongside p-values, as statistical significance alone is insufficient.
- Discuss the practical significance of your findings vs. statistical significance.

5.3 Correlation & Driver Analysis

- Compute a correlation matrix across numerical features and identify the strongest drivers of price.
- Use regression (OLS or similar) to quantify the marginal impact of key predictors on price.
- Apply multicollinearity checks (VIF) and discuss implications for your regression model.
- Identify non-linear relationships using visualizations (scatter plots, LOWESS curves).

5.4 Multi-City Comparisons (if applicable)

Difficulty	Estimated Effort	Category
🟠 Advanced	4–8 Hours	Optional

- Apply appropriate multi-comparison corrections (Bonferroni, FDR) when comparing across many cities.
- Use effect sizes to compare the magnitude of price premiums across different markets.
- Identify markets where statistical patterns diverge significantly from the global trend.

> 📐 **Statistical Methodology Note**
>
> For every statistical test, explicitly document: • Why this test was selected for this data and question • Whether assumptions (normality, independence, equal variance) were verified • How you handled violations of assumptions • How results should be interpreted by a non-technical stakeholder

06 Data Science Challenges

Difficulty	Estimated Effort	Category
🟠 Advanced	10–20 Hours	Optional

This section invites you to apply predictive modeling and machine learning to the Airbnb dataset. These challenges are entirely optional, but candidates who approach them with rigor and depth will demonstrate readiness for senior technical discussions.

6.1 Price Prediction

- Frame the price prediction problem: what is the target variable and how will you evaluate success?
- Perform feature engineering: extract amenity flags, encode categorical variables, create interaction terms.
- Train and compare at least three model families (e.g., Linear, Tree-based, Gradient Boosting).
- Implement proper cross-validation with appropriate metrics (MAE, RMSE, MAPE).
- Analyze residuals: are errors systematic by location, property type, or price range?
- Apply model explainability tools (SHAP, LIME) to identify the most impactful features.

6.2 Demand & Availability Forecasting

- Using calendar data, build a forecast of listing availability over time.
- Identify and model seasonality patterns in booking behavior.
- Discuss the challenges of demand forecasting in a two-sided marketplace.

6.3 Host & Listing Segmentation

- Apply clustering algorithms (K-Means, DBSCAN, or hierarchical) to segment listings into distinct groups.
- Profile each segment: what are the defining characteristics and business implications?
- Segment hosts by behavior patterns: pricing strategy, availability management, response practices.
- Evaluate clustering quality using silhouette scores and business interpretability.

6.4 Model Generalization & Bias

- Test whether your price model generalizes across neighbourhoods and determine whether it favors certain areas?
- If multiple cities: test cross-city model transfer. Train on City A, evaluate on City B.
- Identify potential biases in training data and discuss fairness implications.
- Recommend mitigations for identified biases before production deployment.

> 🔬 **Data Science Documentation Standard**
>
> For each model experiment, document in your report: • Problem framing and success criteria • Feature engineering decisions and rationale • Model selection rationale and hyperparameter choices • Validation

strategy and results • Failure modes and limitations • How you would improve the model with more time or data

07 AI and Machine Learning Opportunities

Difficulty	Estimated Effort	Category
🟣 Expert	8–16 Hours	Optional

Modern data engineering and analytics increasingly intersect with applied AI, from NLP on unstructured data to LLM-powered insight generation and agentic workflows. This section invites bold experimentation.

7.1 Natural Language Processing on Reviews

- Apply sentiment analysis to guest reviews and correlate sentiment with numerical review scores.
- Perform topic modeling (LDA, NMF, BERTopic) on review text to surface recurring themes.
- Extract named entities (landmarks, amenities, complaints) from reviews using NER.
- Build a review quality classifier: distinguish informative reviews from superficial ones.
- Identify language patterns that predict low vs. high review scores.

7.2 LLM-Powered Insight Generation

- Use an LLM (GPT-4o, Claude, Gemini) to generate natural-language summaries of your analytical findings.
- Implement a Retrieval-Augmented Generation (RAG) system over the reviews corpus.
- Build a Q&A interface that allows stakeholders to query the dataset in natural language.
- Generate automated listing improvement recommendations using LLM analysis of review text.

7.3 Recommendation & Discovery Systems

- Design a listing recommendation system based on user preferences and historical data.
- Implement content-based filtering using listing descriptions and amenity vectors.
- Explore collaborative filtering concepts and their applicability to this dataset.
- Discuss cold-start challenges and your proposed solutions.

7.4 Generative AI & Agentic Experimentation

- Design a data analysis agent that autonomously explores the dataset and surfaces insights.
- Experiment with AI-generated listing descriptions and compare them with human-written equivalents.
- Build an AI-powered dynamic pricing advisor that recommends price adjustments based on market signals.
- Propose a responsible AI framework for deploying ML models in an Airbnb-like marketplace.
- Design or prototype an MLOps workflow for continuous model retraining and monitoring.

> 🧠 **Encourage Experimentation**
>
> There is no 'correct' answer in this section. We want to see how you think about AI application design, how you evaluate model outputs critically, and how you document both successes and failures. A well-documented failed experiment is more valuable than an undocumented success.

08 Open Innovation Challenge

Difficulty	Estimated Effort	Category
🟢–🟣 Candidate Defined	Candidate Defined	Optional

This is the section we are most excited about. There are no predefined tasks here. Instead, we challenge you to build something that creates genuine value, demonstrates your unique perspective, or explores territory we have not anticipated.

> ⚡ **Surprise Us**
>
> We are actively looking for originality. The candidates who stand out are those who see possibilities beyond the prescribed tasks and have the initiative to pursue them. Think like a product builder, not just a task executor.

8.1 Inspiration (Not Prescription)

The following are offered as inspiration only. Your own ideas are equally welcome.

Idea	Description
Interactive Dashboard	A live, exploratory dashboard (Streamlit, Dash, Observable, Power BI) for market analysis

Geospatial Visualization	An interactive map-based explorer of market dynamics across neighbourhoods
Production Architecture	A fully designed cloud architecture diagram with cost estimates and trade-off analysis
MLOps Workflow	An automated pipeline from data ingestion through model training to deployment
AI Analyst Assistant	A conversational assistant that answers market questions against the dataset
Stream Processing Sim	Simulate real-time price monitoring and alerting using streaming architecture concepts
Cost Optimization Design	An architectural proposal that reduces processing cost for a global-scale rollout
Responsible AI Framework	A comprehensive framework for ethical AI deployment in rental pricing
Novel Analytical Approach	An analytical lens or methodology not covered anywhere in this document
Automated Reporting	A system that auto-generates a weekly market intelligence PDF for stakeholders

09 City Selection Guide

You have complete freedom in choosing which cities to analyze. The Inside Airbnb dataset covers dozens of cities worldwide across multiple continents. Your selection should be driven by the skills you want to showcase and the scope of analysis you plan to conduct.

City Count	Difficulty	What It Demonstrates
1 City	🟢 Beginner–Intermediate	Focus on depth, code quality, and storytelling. Master one market before generalizing.
2–3 Cities	🔵 Intermediate–Advanced	Introduce comparative analytics, cross-market insights, and multi-source pipeline design.
4–6 Cities	🟠 Advanced	Demonstrate pipeline scalability, automation, configuration-driven processing, and broader market patterns.
7+ Cities	🟣 Expert	Showcase production-grade architecture, performance optimization, and global-scale analytical thinking.

📌 More cities selected → More opportunities to demonstrate scalability, automation, and cross-market insight. However, quality always outweighs quantity. One city analyzed with exceptional depth beats five cities skimmed superficially.

10 AI Tools Usage Policy

Expernetic actively encourages the thoughtful and responsible use of AI tools throughout this assignment. We believe that effective AI collaboration is itself a core professional skill, and we want to see how you leverage these tools while maintaining intellectual ownership of your work.

🤖 **AI Tools Are Encouraged**

You may use any AI tool you find helpful: ChatGPT · Claude · Gemini · GitHub Copilot · Cursor · Perplexity · Local LLMs · Any other

Open-source and free tools are encouraged but not mandatory. Use what makes you most effective.

10.1 Disclosure Requirements

Disclosure Item	What to Include
AI Tools Used	List every AI tool and model version you used during the assignment
AI-Assisted Sections	Clearly identify which parts of your code, analysis, or report were AI-assisted
Prompts Used	Include key prompts in an appendix — show us how you directed the AI
Output Validation	Explain how you verified, tested, and validated AI-generated outputs
Modifications Made	Document meaningful changes you made to AI-generated code or text
Critical Assessment	Note any cases where you rejected or substantially modified AI suggestions

Treating AI disclosure as an afterthought is itself a red flag. We are looking for candidates who approach AI tools with intentionality, critical thinking, and professional accountability.

11 Deliverables

11.1 Mandatory Deliverables

- **Source Code:** All code used in the assignment, organized in a clear directory structure with a README
- **Reproducibility Instructions:** Step-by-step setup guide: dependencies, environment, data download, execution order
- **Professional PDF Report:** Comprehensive written report (see Section 12 for full requirements)
- **Assumptions & Decisions Log:** Documented assumptions made and major technical decisions explained
- **Summary: Completed Work:** Clear summary of what was implemented and to what standard
- **Summary: Incomplete Work:** An honest summary of what was not completed and why, including the rationale behind prioritization decisions.
- **AI Usage Disclosure:** Full AI usage disclosure as specified in Section 10

11.2 Optional Deliverables (High-Value)

- **Jupyter Notebooks:** Annotated notebooks with narrative explaining analytical decisions
- **Interactive Dashboard:** Deployed or locally runnable dashboard application
- **Presentation Deck:** Slide deck summarizing findings for a business audience
- **Demo Video:** Short screen recording walking through your most impressive components
- **Docker Compose:** Containerized environment ensuring full reproducibility
- **Architecture Diagram:** Visual diagram of your data architecture with tool annotations
- **Cloud Deployment:** Live cloud deployment with access URL and brief documentation
- **Unit Tests:** Automated test suite for data quality checks and pipeline components
- **dbt Models:** dbt project with documented models, tests, and lineage graphs

11.3 Submission Instructions

Submit your work via a shared GitHub repository (preferred) or a compressed archive. Ensure your repository README clearly lists all components, how to run them, and the order in which artifacts should be reviewed. Ensure all credentials, API keys, and personal data are excluded or anonymized.

> 📁 Your repository README is your first impression. Treat it with the same care as the rest of your submission.

12 Final Report Requirements

Your report is the primary artifact through which interviewers will evaluate your thinking, communication, and analytical depth. It must be professional, well-structured, and readable by both technical and non-technical stakeholders.

#	Section	Guidance
1	Executive Summary	A 1-page business-oriented summary of your most important findings and recommendations
2	Objectives & Scope	What you set out to achieve, the cities selected, and your prioritization rationale
3	Dataset Overview	Dataset description, files used, relationships, limitations, and assumptions
4	Methodology	Your overall analytical approach and the reasoning behind key method selections
5	Engineering Approach	Pipeline design, architecture decisions, data model, and production considerations
6	EDA Findings	Key patterns, visualizations, and their business interpretations
7	Statistical Findings	Hypothesis results, effect sizes, and statistical conclusions in business terms
8	Data Science Experiments	Model results, comparison tables, validation metrics, and interpretation
9	AI/ML Experiments	NLP, LLM, or agentic work, including methodology, results, and critical evaluation.
10	Visualizations	High-quality figures with numbered captions and clear interpretations
11	Business Recommendations	Actionable recommendations derived from your analysis
12	Cross-City Comparisons	If applicable: comparative insights, regional patterns, and market dynamics
13	Limitations & Caveats	Honest discussion of data limitations, model weaknesses, and scope boundaries
14	Future Improvements	What you would do next with more time, better data, or additional resources

15	Reflection	How you prioritized work, what trade-offs you made, and key lessons learned
Appendix A	AI Usage Disclosure	Full AI tool list, assisted sections, key prompts, and validation notes

> Report Format: PDF, minimum 20 pages. Use professional typography, numbered sections, page numbers, and a table of contents. Visual consistency is expected.

13 Evaluation Rubric

Your submission will be evaluated across the following dimensions. Scores are holistic and qualitative, and this rubric is intended to guide your effort rather than serve as a mechanical checklist.

Evaluation Dimension	Weight	What We Look For
Problem Solving	30	Approach to unfamiliar problems; structured thinking; trade-off awareness
Data Engineering Quality	25	Pipeline design, cleaning rigor, data modeling, production readiness
Statistical Thinking	20	Appropriate test selection, assumption verification, effect size reporting
Analytical Storytelling	20	Business interpretation; narrative clarity; visualization quality
Data Science & ML	15	Feature engineering, model selection, validation, explainability
AI/ML Experimentation	10	NLP, LLM usage, agentic design; critical evaluation of results
Code Quality	20	Readability, modularity, documentation, reproducibility, testing
Communication	20	Report clarity; technical accuracy; accessibility to non-technical readers
Creativity & Initiative	20	Novel approaches; open innovation; going beyond the brief
Prioritization	15	Thoughtful scoping; honest prioritization rationale; quality over quantity

Professionalism	**15**	Attention to detail; file organization; submission quality
AI Usage Disclosure	**10**	Transparency; critical evaluation of AI outputs; responsible practice

⭐ **Important Evaluation Note**

Candidates who demonstrate thoughtful prioritization, strong reasoning, depth in selected areas, and clear communication may outperform candidates who attempt many tasks superficially.

Candidates who successfully explore multiple cities, multiple analytical disciplines, and greater scope may earn additional recognition for handling increased complexity, provided quality is maintained.

14 Technical Toolkit Guidance

There are no mandatory tools or languages. We encourage you to use the stack you are most proficient in. The following table lists commonly used tools across each category and should be treated as inspiration rather than a prescribed set of requirements.

Category	Suggested Tools (not mandatory)
Languages	Python (preferred), SQL, R, Scala, Julia
Data Processing	pandas, polars, PySpark, DuckDB, dbt, SQL
Visualization	matplotlib, seaborn, plotly, Altair, Vega-Lite, Power BI, Tableau
ML & Statistics	scikit-learn, XGBoost, LightGBM, statsmodels, scipy, PyTorch
NLP & LLMs	spaCy, NLTK, HuggingFace Transformers, LangChain, OpenAI, Anthropic SDK
Geospatial	GeoPandas, Folium, Kepler.gl, QGIS, deck.gl
Orchestration	Prefect, Airflow, Luigi, dbt
Databases	PostgreSQL, DuckDB, SQLite, BigQuery, Snowflake
Cloud Platforms	AWS (S3, Glue, Athena), GCP (BigQuery, Dataflow), Azure (ADLS, Synapse)
Containers	Docker, docker-compose, Kubernetes (optional)
Dashboards	Streamlit, Dash, Observable, Gradio, Metabase
Version Control	Git, GitHub (required for submission)
AI Coding Tools	GitHub Copilot, Cursor, Claude, ChatGPT, Codeium